HALL'S THEOREMS ON SOLVABLE GROUPS

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Abstract. Representation theory is a way of studying the structure of groups using the tools and ideas of linear algebra. The goal of this paper is to demonstrate the power of representation theory to derive nontrivial properties about group structure. The culminating result of this paper will be a proof of all of Hall's theorems regarding solvable groups. Hall's theorems extend the existence, conjugacy, and containment statements of the Sylow theorems from a single prime to an arbitrary set of primes, under the additional assumption that the group is solvable.

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1. Introduction

Representation theory was invented by Frobenius to understand a bizarre pattern. If one takes the Cayley table of a group and considers each element as a variable x_g, then the determinant

$$\Theta_G = \det(x_{gh^{-1}})_{g,h\in G}$$

is a product of irreducible polynomials, such that the exponent on each factor is equal to its degree. Representation theory seeks to use the tools and insights of linear algebra to study a wide range of diverse mathematical structures by "representing" them in a form that is easier to understand. This turns out to be an extremely fruitful endeavor, and today representation theory is extremely widespread. In this paper, I will define the basic notions and results of representation theory as well as character theory. Finally, I will use these results to provide a complete treatment of Hall's theorems regarding solvable groups. Along the way, I will use representation theory to prove Burnside's theorem, as well as a weakened version of Schur-Zassenhaus, both of which are extremely famous and powerful results.

2. Representation Theory

Throughout this paper, we will assume the base field to be $\mathbb{C}$, all groups are finite, and all vector spaces are finite dimensional. Furthermore, if g is an element of some group which acts linearly on a vector space V via a group action, I will denote $g|_V \in \mathrm{End}(V)$ as the linear transformation defined by g. We start by fixing a finite group G.

2.1. **Basic Representation Theory.**

Definition 2.1. The *group algebra* $\mathbb{C}[G]$ is the set of all linear combinations of elements in G with coefficients in $\mathbb{C}$. Multiplication and addition are defined in the natural way:

$$\sum_{g\in G} z_g g + \sum_{g\in G} z'_g g = \sum_{g\in G}(z_g + z'_g)g$$

$$\left(\sum_{g\in G} z_g g\right)\left(\sum_{g\in G} z'_g g\right) = \sum_{g,g'\in G}(z_g z'_{g'})(gg')$$

Definition 2.2. A *(complex) representation* of G consists of the data of a pair (ρ, V), where V is a finite dimensional $\mathbb{C}$-vector space and $\rho : G \to \mathrm{GL}(V)$ is a group homomorphism from G to the automorphism group of V. Using ρ, we give an action of $\mathbb{C}[G]$ on V as:

$$\left.\left(\sum_{g\in G} z_g g\right)\right|_V := \sum_{g\in G} z_g \rho(g)$$

- A *subrepresentation* $(\rho|_W, W)$ of (ρ, V) consists of the data of a vector subspace W of V and group homomorphism $\rho|_W : G \to \mathrm{GL}(W)$ given by restriction of ρ to W, i.e. for any $g \in G$, we have $\rho|_W(g) = \rho(g)|_W$.
- We say that the representation (ρ, V) is *irreducible* if it only has trivial subrepresentations, i.e. the zero representation $(\rho|_0, 0)$ and (ρ, V) itself.

One may think of a group representation more simply as a $\mathbb{C}$-linear action of $\mathbb{C}[G]$ on V. For this reason, we sometimes only denote V for the representation (ρ, V). In this way, we can view a subrepresentation W of V as a vector subspace which is stable under the action of $\mathbb{C}[G]$. We often don't need to invoke the larger structure of the group algebra, and in many of our proofs we will only consider the actions of elements of G.

Example 2.3. Consider the Klein four group $\langle a, b | a^2 = b^2 = (ab)^2 = e\rangle$. It has this representation:

$$a \mapsto \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \qquad b \mapsto \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

This representation has a subrepresentation which consists of the vector subspace of $\mathbb{C}^2$ spanned by $(1, 0)$. In this subrepresentation our map looks like:

$$a \mapsto -1 \qquad b \mapsto 1$$

This representation is trivially irreducible because it is one dimensional and therefore has no nontrivial vector subspaces.

Lemma 2.4. *If V is a representation of G over $\mathbb{C}$ and $g \in G$, we have the following lemmas:*

(a) $e|_V = \mathrm{Id}$
(b) $g|_V$ *is diagonalizable.*
(c) The eigenvalues of $g|_V$ *are roots of unity.*

Proof.

(a) $\rho : G \to \mathrm{GL}(V)$ is a group homomorphism hence it must map the identity to the identity.
(b) Since G is a finite group, g has finite order and we must have ${g|_V}^{|g|} = \mathrm{Id}$. Then $g|_V$ satisfies the polynomial equation $X^{|g|} - 1 = 0$, which has no repeated roots. The minimal polynomial of $g|_V$ must divide this polynomial, and therefore it also has no repeated roots. Therefore $g|_V$ is diagonalizable.
(c) Since the roots of $X^{|g|} - 1$ are the $|g|$-th roots of unity and the minimal polynomial of $g|_V$ divides $X^{|g|} - 1$, the eigenvalues of $g|_V$ must be $|g|$-th roots of unity.

□

Now that we have defined what a representation is, and identified a particularly useful type of representation (irreducible representations), we will study maps between representations.

Definition 2.5. Given two representations V and W, an *intertwiner* between them is a linear operator $\phi : V \to W$ which commutes with the action of G, i.e. for any $v \in V$ and $g \in G$, $\phi(gv) = g\phi(v)$. The set of intertwiners from V to W is denoted $\mathrm{Hom}_G(V, W)$.

If ϕ is an isomorphism of vector spaces then it is an *isomorphism of representations*. Two representations are *isomorphic* if there is an isomorphism between them.

The following result, Schur's Lemma, is extremely powerful and fundamental to the study of representation theory.

Theorem 2.6 (Schur's Lemma)**.** *Suppose V and W are representations of G and $\phi : V \to W$ is a nonzero intertwiner between them. Then:*

(a) If V is irreducible then ϕ is injective.
(b) If W is irreducible then ϕ is surjective.
(c) If $W = V$ is irreducible then $\phi = \lambda Id$ for some $\lambda \in \mathbb{C}$.

Proof. We use the defining property of irreducible representations.

(a) Consider the subspace $\ker(\phi)$ of V. Because ϕ commutes with the action of G, $\ker(\phi)$ is invariant under the action of G and hence is a subrepresentation of V. Therefore, since V is irreducible and ϕ is nonzero, we must have $\ker(\phi) = 0$ and ϕ is injective.

(b) Similarly, consider the subspace $\operatorname{im}(\phi)$ of W. Because ϕ commutes with the action of G, $\operatorname{im}(\phi)$ is invariant under the action of G and is a subrepresentation of W. Therefore, since W is irreducible and ϕ is nonzero, we must have $\operatorname{im}(\phi) = W$ and ϕ is surjective.

(c) Since $\mathbb{C}$ is algebraically closed we may choose an eigenvalue λ of ϕ. Note that $\phi - \lambda \operatorname{Id}$ is singular. If $\phi - \lambda \operatorname{Id}$ is nonzero, then by (a) $\phi - \lambda \operatorname{Id}$ is injective, which contradicts being singular. Therefore $\phi - \lambda \operatorname{Id} = 0$, i.e. $\phi = \lambda \operatorname{Id}$.

□

Corollary 2.7. *If V is an irreducible representation of G and $\alpha \in Z(\mathbb{C}[G])$, then $\alpha|_V = \lambda \operatorname{Id}$ for some $\lambda \in \mathbb{C}$.*

Proof. Since α lies in the center of $\mathbb{C}[G]$ its action on V is an intertwiner $V \to V$. The result follows from Schur's lemma. □

Now we will generalize Schur's lemma to a more powerful form.

Lemma 2.8. *If $A = \bigoplus_i V_i^{\oplus n_i}$ and $B = \bigoplus_i V_i^{\oplus r_i}$ are two direct sums of pairwise distinct irreducible representations V_i with multiplicities n_i and r_i respectively, then:*

$$\operatorname{Hom}_G(A, B) \cong \bigoplus_i M_{r_i \times n_i}(\mathbb{C})$$

Proof. By the universal properties of direct sums, the space of intertwiners decomposes bi-additively:

$$\operatorname{Hom}_G(A, B) \cong \bigoplus_{i,k} \operatorname{Hom}_G\left(V_i^{\oplus n_i}, V_k^{\oplus r_k}\right)$$

By Schur's Lemma, $\operatorname{Hom}_G(V_i, V_k) = 0$ whenever $i \neq k$, since the representations are distinct and irreducible. Thus, all terms with $i \neq k$ vanish, leaving:

$$\operatorname{Hom}_G(A, B) \cong \bigoplus_i \operatorname{Hom}_G\left(V_i^{\oplus n_i}, V_i^{\oplus r_i}\right)$$

For a fixed i, an intertwiner $\varphi_i \in \operatorname{Hom}_G(V_i^{\oplus n_i}, V_i^{\oplus r_i})$ is completely determined by its component maps $\varphi_{kl} : V_i \to V_i$ for $1 \leq l \leq n_i$ and $1 \leq k \leq r_i$. Applying Schur's Lemma again over $\mathbb{C}$, each endomorphism of V_i is a scalar multiple of the identity

$$\varphi_{kl} = \lambda_{kl} \operatorname{Id}_{V_i} \quad \text{for some } \lambda_{kl} \in \mathbb{C}.$$

The assignment $\varphi_i \mapsto (\lambda_{kl})$ defines a canonical vector space isomorphism between $\operatorname{Hom}_G\left(V_i^{\oplus n_i}, V_i^{\oplus r_i}\right)$ and the matrix space $M_{r_i \times n_i}(\mathbb{C})$. Summing over all irreducible components i gives:

$$\operatorname{Hom}_G(A, B) \cong \bigoplus_i M_{r_i \times n_i}(\mathbb{C})$$

□

2.2. **Irreducible Representations.** Now we will prove some theorems which should establish the importance of the irreducible representations. In particular, we will eventually show that arbitrary representations factor uniquely (up to permutation) into a direct sum of irreducible representations (Krull-Schmidt theorem). In this way, irreducible representations are analogous to the prime numbers of representation theory.

Remark 2.9. If V and W are representations of G, we can naturally consider $\mathrm{Hom}(V,W)$ as a representation of G. For $f \in \mathrm{Hom}(V,W)$, $v \in V$ and $g \in G$ we define

$$(g \cdot f)(v) = g \cdot f(g^{-1} \cdot v)$$

which is a representation since:

$$(g \cdot (h \cdot f))(v) = g \cdot ((h \cdot f)(g^{-1}v)) = (gh) \cdot f(h^{-1}g^{-1}v) = ((gh) \cdot f)(v)$$

From the definition it is clear that some $f \in \mathrm{Hom}(V,W)$ is fixed by the action of G if and only if it is an intertwiner $V \to W$. This gives

$$\mathrm{Hom}_G(V,W) = \mathrm{Hom}(V,W)^G$$

where $\mathrm{Hom}(V,W)^G$ denotes the G fixed points of $\mathrm{Hom}(V,W)$.

Lemma 2.10. *If V is a representation of G and W is a subrepresentation of V, there exists a surjective projection $\pi : V \to W$ which is an intertwiner.*

Proof. Take an arbitrary surjective projection $\pi_0 : V \to W$. I claim that

$$\pi := \frac{1}{|G|} \sum_{g \in G} g \cdot \pi_0$$

is the desired map. There are a few things to check.

- It is an intertwiner since for any $v \in V$ and $x \in G$

$$x \cdot \pi = \frac{1}{|G|} \sum_{g \in G} (xg) \cdot \pi_0 = \frac{1}{|G|} \sum_{x^{-1}g \in G} g \cdot \pi_0 = \pi$$

  so $\pi \in \mathrm{Hom}(V,W)^G = \mathrm{Hom}_G(V,W)$.
- Its image lies in W since for any $v \in V$

$$\pi(v) = \frac{1}{|G|} \sum_{g \in G} g\pi_0(g^{-1}v)$$

  is a sum of elements of W since π_0 projects onto W and W is G-stable.
- It acts by identity on W since for any $w \in W$

$$\pi(w) = \frac{1}{|G|} \sum_{g \in G} g\pi_0(g^{-1}w) = \frac{1}{|G|} \sum_{g \in G} gg^{-1}w = w$$

  because $\pi_0(g^{-1}w) = g^{-1}w$ by the fact that π_0 fixes W.

□

Theorem 2.11 (Maschke's Theorem + Krull-Schmidt Theorem)**.** *Any representation decomposes uniquely into a direct sum of irreducible representations.*

Proof. Suppose V is a representation of G. We first prove that V decomposes into a direct sum of irreducible representations (Maschke's theorem), and then prove any two such decompositions are unique up to multiplicities of irreducible isomorphism classes (Krull-Schmidt theorem).

Maschke's Theorem: This is by induction on the dimension of V. If V is irreducible we are done. Otherwise, V must have a subrepresentation $W \subset V$. By Lemma 2.10 we may find a surjective projection $\pi : V \to W$ which is an intertwiner. Define $W' := \ker(\pi)$. Since π is an intertwiner W' is G-invariant. By construction $V = W \oplus W'$. Now the inductive hypothesis applies to W and W'.

Krull-Schmidt Theorem: Suppose that $V = \bigoplus_i V_i^{n_i} = \bigoplus_i V_i^{r_i}$ are two decompositions of V into distinct irreducible representations V_i with multiplicity. Consider the identity map $\bigoplus_i V_i^{n_i} \to \bigoplus_i V_i^{r_i}$. By Lemma 2.8 this map looks like $\bigoplus_i M_i$ where each $M_i : V_i^{n_i} \to V_i^{r_i}$ is a $r_i \times n_i$ matrix. This map is invertible and therefore $r_i = n_i$ for each i.

□

Lemma 2.12. *If $V = V_1^{p_1} \oplus \cdots \oplus V_n^{p_n}$ is a sum of distinct irreducible representations (with multiplicity) and $W \subseteq V$ is a subrepresentation of V then W is isomorphic to $V_1^{r_1} \oplus \cdots \oplus V_n^{r_n}$ where $r_i \leq p_i$.*

Proof. Consider W' as in the proof of Maschke's theorem. Note that W and W' are both subrepresentations of V with $V = W \oplus W'$. The result follows from the Krull-Schmidt theorem. □

Hopefully, these theorems have provided some insight as to why we care so much about the irreducible representations. I will prove one more useful theorem which is closely related to these.

Theorem 2.13 (Density Theorem)**.** *If V is an irreducible representation of G the map $\rho : \mathbb{C}[G] \to \mathrm{End}(V)$ is surjective.*

Proof. Choose an arbitrary basis $v_1, \ldots, v_n$ of V and consider the map $\alpha : \mathbb{C}[G] \to V^n$ as $\alpha(a) = (av_1, \ldots, av_n)$. Since an endomorphism is determined by its action on a set of basis vector, it suffices to prove that this map is surjective. Suppose the contrary. Then the image of α is a proper subrepresentation of V^n and by Lemma 2.12 we have $\mathrm{im}(\alpha) \cong V^r$ where $r < n$. Let $\phi : V^r \to V^n$ be the inclusion map. By Lemma 2.8 ϕ looks like an $n \times r$ matrix. Therefore any $(y_1, \ldots, y_n) = \phi(x_1, \ldots, x_r)$ in the image has each y_i a linear combination of $x_1, \ldots, x_r \in V$. Note that $\alpha(e) = (v_1, \ldots, v_n)$, therefore $(v_1, \ldots, v_n)$ is in the image of ϕ. This is a contradiction since $v_1, \ldots, v_n$ are linearly independent, and cannot be simultaneously in the span of r vectors. □

2.3. **Class Functions and Characters.**

Now that we have introduced irreducible representations, a natural question that arises is how many irreducible representations can be found for a given group? In this section, we will prove that the number of irreducible representations a group has (up to isomorphism) is equal to the number of conjugacy classes it contains.

Definition 2.14. Given a group G, a *class function* $\alpha : G \to \mathbb{C}$ is a function which is constant on conjugacy classes. Specifically for each $a, b \in G$, $\alpha(aba^{-1}) = \alpha(b)$.

Note that the space of class functions on G is a vector space over $\mathbb{C}$ whose dimension equals the number of conjugacy classes of G.

Definition 2.15. Given a representation V of a group G, we define the *character* $\chi_V : G \to \mathbb{C}$ of V as $\chi_V(a) = \mathrm{tr}\, a|_V$. This is a class function because the trace is invariant under conjugation.

Definition 2.16. We define a Hermitian inner product on the vector space of class functions as $\langle \phi, \psi \rangle = \frac{1}{|G|} \sum_{g \in G} \phi(g)\overline{\psi(g)}$.

As we will see, this is the most useful and natural inner product on the space of class functions. We will utilize its nice properties extensively throughout the rest of the paper.

Theorem 2.17. *If V and W are representations of G then:*

$$\langle \chi_V, \chi_W \rangle = \dim \operatorname{Hom}_G(V, W)$$

Proof. Since G acts linearly on $\operatorname{Hom}(V, W)$ by Remark 2.9 we can consider the trace of $g|_{\operatorname{Hom}(V,W)}$. Fix an eigenbasis $v_1, \ldots, v_n$ of V and $w_1, \ldots, w_r$ of W with respect to the $g|_V$ and $g|_W$ respectively. Let their eigenvalues be $\lambda_{v_1}, \ldots, \lambda_{v_n}$ and $\lambda_{w_1}, \ldots, \lambda_{w_r}$. We construct a basis of $\operatorname{Hom}(V, W)$ as the transformations which take an eigenvector of $g|_V$ to an eigenvector of $g|_W$ and kill all others. Then $g|_{\operatorname{Hom}(V,W)}$ is a diagonal matrix whose trace we can calculate:

$$\begin{aligned} \operatorname{tr} g|_{\operatorname{Hom}(V,W)} &= \sum_{i,j} \frac{\lambda_{w_i}}{\lambda_{v_j}} \\ &= \sum_{i,j} \lambda_{w_i} \overline{\lambda_{v_j}} && \textit{Lemma } 2.4 \\ &= \left(\sum_i \lambda_{w_i} \right) \left(\sum_j \overline{\lambda_{v_j}} \right) \\ &= \operatorname{tr} g|_W \overline{\operatorname{tr} g|_V} \end{aligned}$$

Now consider the linear transformation $\operatorname{Hom}(V, W) \to \operatorname{Hom}(V, W)$:

$$\alpha := \frac{1}{|G|} \sum_{g \in G} g|_{\operatorname{Hom}(V,W)}$$

Noting that $\operatorname{Hom}_G(V, W) = \operatorname{Hom}(V, W)^G$, a short computation yields that:

- α fixes $\operatorname{Hom}_G(V, W)$.
- The image of α is contained in $\operatorname{Hom}_G(V, W)$.

Therefore α is a projection from $\operatorname{Hom}(V, W)$ onto $\operatorname{Hom}_G(V, W)$ and $\dim \operatorname{Hom}_G(V, W) = \operatorname{tr} \alpha$. Now:

$$\begin{aligned} \operatorname{tr} \alpha &= \frac{1}{|G|} \sum_{g \in G} \operatorname{tr} g|_{\operatorname{Hom}(V,W)} \\ &= \frac{1}{|G|} \sum_{g \in G} \operatorname{tr} g|_W \overline{\operatorname{tr} g|_V} \\ &= \frac{1}{|G|} \sum_{g \in G} \chi_W(g) \overline{\chi_V(g)} \\ &= \langle \chi_W, \chi_V \rangle \end{aligned}$$

Since $\dim \operatorname{Hom}_G(V, W)$ is real we can swap the terms in the inner product and write

$$\langle \chi_V, \chi_W \rangle = \dim \operatorname{Hom}_G(V, W)$$

as desired. □

This idea of averaging over the action of G to create a G invariant map is an extremely powerful problem solving technique. It is probably the single most useful idea in this whole paper.

Corollary 2.18 (First Orthogonality Relation)**.** *The characters of irreducible representations are orthonormal with respect to the outlined inner product.*

Proof. Let V and W be irreducible representations. Note that

$$\langle \chi_V, \chi_W \rangle = \dim \mathrm{Hom}_G(V, W)$$

and by Schur's lemma

$$\dim \mathrm{Hom}_G(V, W) = \begin{cases} 1 & V \cong W \\ 0 & \text{otherwise} \end{cases}$$

proving the theorem ☐

An important corollary of this theorem is that there are no more irreducible representations than conjugacy classes of G. This is because orthogonal vectors are linearly independent, and we cannot have more linearly independent vectors than the dimension of the space. You might ask if this bound is attained, that is, are there exactly as many distinct irreducible representations as conjugacy classes of G. We will soon see the answer is yes.

2.4. **The Regular Representation.**

We now introduce the regular representation. Just as examining the action of G on itself is often useful in group theory, examining the action of $\mathbb{C}[G]$ on itself proves extremely useful in representation theory.

Definition 2.19 (Regular Representation)**.** There is the natural $\mathbb{C}$-linear action of G on $\mathbb{C}[G]$ defined for any $x \in G$ by:

$$x \cdot \left(\sum_{g \in G} z_g g \right) = \sum_{g \in G} z_g (xg)$$

This defines the *left regular representation* of G.

We will refer to the left regular representation as simply the regular representation in this paper.

Theorem 2.20. *The characters of the irreducible representations form a basis of the vector space of class functions.*

Proof. Suppose the contrary, then there is some nonzero class function ϕ with $\langle \chi_V, \phi \rangle = 0$ for all irreducible representations. Define

$$\alpha = \frac{1}{|G|} \sum_{g \in G} \overline{\phi(g)} g$$

and consider an arbitrary irreducible representation V. Since ϕ is a class function α lies in the center of $\mathbb{C}[G]$ and therefore $\alpha|_V = \lambda \,\mathrm{Id}$ for some $\lambda \in \mathbb{C}$ by Corollary 2.7.

However note that

$$\begin{aligned}\operatorname{tr}\alpha|_V &= \frac{1}{|G|}\sum_{g\in G}\overline{\phi(g)}\chi_V(g)\\ &= \langle \chi_V, \phi\rangle\\ &= 0\end{aligned}$$

hence $\alpha|_V = 0$. Therefore by Maschke's theorem α acts by 0 on any representation, including the regular representation $\mathbb{C}[G]$. Since the elements of G are linearly independent in $\mathbb{C}[G]$, $\alpha|_{\mathbb{C}[G]} = 0$ implies $\phi(g) = 0$ for each $g \in G$. However we stated that ϕ was nonzero, a contradiction. □

Corollary 2.21. *The number of irreducible representations equals the number of conjugacy classes.*

2.5. **The Character Table.** Now that we have proven Corollary 2.21, an extremely important result, we have enough machinery to define a very important concept. That is the character table, a way to encode all the relevant information about the characters of a group's irreducible representations. Note that $C_G(g)$ denotes the centralizer of g.

For the remainder, if C is a conjugacy class of G and ϕ is a class function, we denote $\phi(C) := \phi(c)$ for any $c \in C$.

Definition 2.22. The *character table* X of a group is the matrix whose rows correspond to irreducible representations and whose columns correspond to conjugacy classes. If the irreducible representations are $V_1, \ldots, V_n$ and the conjugacy classes are $C_1, \ldots, C_n$ then $X_{ij} = \chi_{V_i}(C_j)$. Note that the character table is only defined up to permutation of rows and columns.

Example 2.23. The character table of the Klein four group $\langle a, b | a^2 = b^2 = (ab)^2 = e\rangle$ is as follows:

	e	a	b	ab
χ_1	1	1	1	1
χ_2	1	1	-1	-1
χ_3	1	-1	1	-1
χ_4	1	-1	-1	1

Theorem 2.24 (Second Orthogonality Relation)**.** *Suppose $h, g \in G$, then:*

$$\sum_{V\in\mathrm{Irr}}\chi_V(g)\overline{\chi_V(h)} = \begin{cases}|C_G(g)|, & \textit{if } g \textit{ and } h \textit{ are conjugate}\\ 0, & \textit{otherwise}\end{cases}$$

where the sum is over the irreducible representations of G.

Proof. Let $C_1, \ldots, C_n$ be the conjugacy classes of our group. By the first orthogonality relation the character table X has orthonormal rows under our inner product.

Using Corollary 2.18 and the definition of our inner product gives:

$$\begin{aligned}
\mathrm{Id} &= \begin{bmatrix} 1 & 0 & \cdots \\ 0 & 1 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} \\
&= \begin{bmatrix} \langle \chi_{V_1}, \chi_{V_1} \rangle & \langle \chi_{V_1}, \chi_{V_2} \rangle & \cdots \\ \langle \chi_{V_2}, \chi_{V_1} \rangle & \langle \chi_{V_2}, \chi_{V_2} \rangle & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} \\
&= X \begin{bmatrix} |C_1|/|G| & 0 & \cdots \\ 0 & |C_2|/|G| & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} X^\dagger \\
X^\dagger X &= \begin{bmatrix} 1/|C_G(c_1)| & 0 & \cdots \\ 0 & 1/|C_G(c_2)| & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}^{-1} \qquad \text{after choosing some } c_i \in C_i
\end{aligned}$$

But $(X^\dagger X)_{ij} = \sum_{V \in \mathrm{Irr}} \chi_V(C_i)\overline{\chi_V(C_j)}$. This proves the result. □

3. Burnside's Theorem

Now we will prove a big result, Burnside's theorem, which states that any group of order $p^\alpha q^\beta$ is solvable. This theorem is the premier demonstration of the power of representation theory to understand the structure of finite groups.

Theorem 3.1 (Burnside's Theorem)**.** *All groups of order $p^\alpha q^\beta$ for primes p, q are solvable.*

We proceed by contradiction, suppose that G is the smallest order nonsolvable group of order $p^\alpha q^\beta$. Notice that G must be a simple group. Otherwise there is some normal subgroup N of G. Then G/N and N are solvable by minimality which implies G is solvable, a contradiction.

We claim that G cannot have any conjugacy class of order p^k or q^k. Writing the class equation for G yields

$$|G| = |Z(G)| + \sum_{C \neq \{g\}} |C|$$

where the sum is over the non-singleton conjugacy classes. From the claim, every non-singleton conjugacy class of G has order divisible by pq. This will imply that:

$$|Z(G)| = 0 \pmod{pq}$$

However $|Z(G)| \geq 1$ because the identity lies in the center. This shows that the center of G is nontrivial. It is also proper since otherwise the group is solvable. However the center is always a normal subgroup, a contradiction with the simplicity of G.

Now we use the following steps to prove that G does not have conjugacy class of prime power order (Theorem 3.6).

Lemma 3.2. *Suppose that $\omega_1, \ldots, \omega_n$ are roots of unity. If $\frac{1}{n}\sum \omega_i$ is an algebraic integer either $\omega_1 = \cdots = \omega_n$ or $\sum \omega_i = 0$*

Proof. Let $\alpha = \frac{1}{n}\sum \omega_i$ be an algebraic integer with $\omega_1, \ldots, \omega_n$ not satisfying $\omega_1 = \cdots = \omega_n$. We take the field norm of α over some cyclotomic field F containing all the ω_i:

$$N_{F/\mathbb{Q}}(\alpha) = \prod_{\sigma \in Gal(F/\mathbb{Q})} \sigma(\alpha)$$

Because α is an algebraic integer $N_{F/\mathbb{Q}}(\alpha)$ must be an integer. If the ω_is are not all the same this implies $|\sigma(\alpha)| < 1$ for each σ by the triangle inequality. Therefore $|N_{F/\mathbb{Q}}(\alpha)| < 1$ and $N_{F/\mathbb{Q}}(\alpha) = 0$. This forces $\alpha = 0$. □

Lemma 3.3. *Suppose V is an irreducible representation of G and C is a conjugacy class of G. Then $\frac{|C|\chi_V(C)}{\dim V}$ is an algebraic integer.*

Proof. Consider the conjugacy class sum $\alpha = \sum_{c \in C} c$, which lies in the center of $\mathbb{C}[G]$. By Corollary 2.7 $\alpha|_V = \lambda \operatorname{Id}$ for some λ. Because $\mathbb{Z}[G]$ is finitely generated over $\mathbb{Z}$, the center of $\mathbb{Z}[G]$ is integral over $\mathbb{Z}$. Therefore $\alpha \in Z(\mathbb{Z}[G])$ is integral over $\mathbb{Z}$ and λ is an algebraic integer. Taking the trace of α yields

$$|C|\chi_V(C) = \lambda \dim V$$
$$\frac{|C|\chi_V(C)}{\dim V} = \lambda$$

and we are done. □

Lemma 3.4. *Suppose V is an irreducible representation of G and C is a conjugacy class of G with $gcd(|C|, \dim V) = 1$. Then either $\chi_V(C) = 0$ or $c|_V = \lambda_C \operatorname{Id}$ for every $c \in C$.*

Proof. Since $\gcd(|C|, \dim V) = 1$, Bézout's lemma gives integers x and y with:

$$x\frac{|C|\chi_V}{\dim V} + y\chi_V(C) = \frac{\chi_V(C)}{\dim V}$$

By Lemma 3.3 $\frac{|C|}{\dim V}\chi_V(C)$ is an algebraic integer. Moreover $\chi_V(C)$ is a sum of roots of unity by Lemma 2.4 and is an algebraic integer. Therefore $\frac{\chi_V(C)}{\dim V}$, being a sum of algebraic integers, is an algebraic integer. Choose any $c \in C$ and write

$$\frac{\chi_V(C)}{\dim V} = \frac{1}{\dim V}(\omega_1 + \omega_2 + \cdots + \omega_{\dim V})$$

where ω_i are the eigenvalues of $c|_V$, which are roots of unity. By Lemma 3.2 either $\omega_1 = \cdots = \omega_{\dim V}$ and $c|_V = \frac{\chi_V(C)}{\dim V} \operatorname{Id}$ or $\sum \omega_i = 0$ and $\chi_V(C) = 0$. □

Lemma 3.5. *Suppose G is a group with a conjugacy class C of order p^k for some prime p and $k \geq 1$. Then there exists a nontrivial irreducible representation V with $\chi_V(C) \neq 0$ whose dimension is not divisible by p.*

Proof. Denote S as the set of irreducible representations whose dimension is divisible by p. Denote T as the set of nontrivial irreducible representations whose dimension is not divisible by p.

Let c be an element in the conjugacy class C. Note for each $V \in S$ that $\frac{1}{p}\dim V \chi_V(c)$ is an algebraic integer. Then

$$\alpha = \frac{1}{p}\sum_{V \in S} \dim V \chi_V(c)$$

is an algebraic integer. Using the second orthogonality relation on the elements e and c gives

$$\sum_{V \in \mathrm{Irr}} \dim V \chi_V(C) = 0$$

because they do not lie in the same conjugacy class. Splitting up the sum

$$\begin{aligned} 0 &= \chi_{\mathbb{C}}(c) + \sum_S \dim V \chi_V(C) + \sum_T \dim V \chi_V(c) \\ &= 1 + p\alpha + \sum_T \dim V \chi_V(C) \end{aligned}$$

where $\chi_{\mathbb{C}}$ denotes the character of the trivial representation. If the sum over T vanishes, then we must have $\alpha = -1/p$. However $-1/p$ is not an algebraic integer, a contradiction. Therefore $\sum_T \dim V \chi_V(C)$ must be nonzero and there is at least one $V \in T$ with $\chi_V(C) \neq 0$. □

Theorem 3.6. *If G is simple, it has no conjugacy classes of order p^k where p is a prime and $k \geq 1$.*

Proof. Suppose the contrary and let C be a conjugacy class of order p^k. By Lemma 3.5, pick a nontrivial irreducible representation V whose dimension is not divisible by p and with $\chi_V(C) \neq 0$. By Lemma 3.4, each $c \in C$ acts on V by scaling by some λ. Consider the following subgroup of G:

$$H := \{g \in G \mid g|_V = \mathrm{Id}\}.$$

We claim that H is a proper nontrivial normal subgroup of G.

- Since V is nontrivial H is a proper subgroup.
- For any $g \in G$ and $h \in H$ $(ghg^{-1})|_V = g|_V \,\mathrm{Id}\, g|_V^{-1} = \mathrm{Id}$, hence $ghg^{-1} \in H$. Therefore H is normal.
- Choose $a \neq b \in C$ and note that

  $$ab^{-1}|_V = \lambda\lambda^{-1} = \mathrm{Id}$$

  so $ab^{-1} \in H$, proving that H is nontrivial.

This is a contradiction with the simplicity of G. □

This is the claim we needed in the beginning, so we are done.

4. Schur-Zassenhaus

The Schur-Zassenhaus theorem is a deep result in finite group theory. It allows us to determine in certain cases that a group can be constructed via a semidirect product. We state the full theorem for context, however we will not prove it.

Theorem 4.1 (Schur-Zassenhaus)**.** *If $|G| = ab$ where a and b are coprime, and N is a normal subgroup of G with order a, then G contains a subgroup of order b. Furthermore, all subgroups of order b are conjugate.*

In this paper, we will assume that N is an elementary abelian p-group (a group in which every nonidentity element has order p). We only need this weaker version to prove Hall's theorems. Nonetheless, one may find a proof that the general case reduces to our special case in Kargapolov and Merzljakov [6] (this requires Feit-Thompson theorem).

Definition 4.2. We call E an *extension* of B by A if it fits into an exact sequence:

$$1 \to A \to E \to B \to 1$$

Equivalently, E has a normal subgroup isomorphic to A, and $E/A \cong B$.

We wish to show that under the specified conditions the exact sequence

$$1 \to N \xrightarrow{\iota} G \xrightarrow{\pi} G/N \to 1$$

splits. This means that G has a subgroup H which is isomorphic to G/N, and such that each element of H lies in its own coset of G/N. Equivalently, there is a group homomorphism $\rho : G/N \to G$ with $\pi \circ \rho = \mathrm{Id}_{G/N}$. This subgroup will have order $|G/N| = b$.

In the above formulation of Schur-Zassenhaus, we want to prove something about G using only the properties of N and G/N (that is, they have coprime order). Effectively, we need to prove something about *every* extension of N by G/N. Therefore we may phrase the theorem we desire to prove in the following way.

Theorem 4.3 (Weak Schur-Zassenhaus)**.** *Suppose that*

$$1 \to A \xrightarrow{\iota} E \xrightarrow{\pi} B \to 1$$

is an exact sequence of finite groups, where A is an elementary abelian p-group and $p \nmid |B|$. Then the exact sequence splits. Moreover any two such splitting maps $\rho_1, \rho_2 : B \to E$ are conjugate.

The advantage of this formulation is that it gives us more direction. We need to try to understand and classify the space of all extensions.

Definition 4.4. Let A and B be finite groups. We denote

$$A^B := \{\text{set-theoretic functions } B \to A\}$$

and note that A^B is a group under pointwise multiplication of functions $(f \cdot g)(b) = f(b) \cdot g(b)$. For each $f \in A^B$ and $b \in B$ we define f^b as

$$f^b(x) = f(b^{-1}x)$$

so that each $b \in B$ defines an automorphism on A^B. This gives us an action of B on A^B. We now define $W = A \wr B$ the *wreath product* of A and B as the semidirect product $A^B \rtimes B$ with the action of B on A^B as defined. More explicitly, $A \wr B$ has the underlying set $A^B \times B$ with group operation given by:

$$fb \cdot f'b' = ff'^b bb'$$

The intuitive interpretation is the following. To each element of B, I associate a copy of A.

- A can act on any of the $|B|$ copies of itself by left multiplication.
- B acts on the set by permuting the copies of A via left multiplication on itself.

The wreath product is the group describing all the actions I may take on this set if I am allowed these two operations. For instance, if R is the group of a Rubik's cube, and C_2 is the cyclic group of order 2, then $R \wr C_2$ is the group of two Rubik's cubes, consisting of all combinations of playing with them and possibly swapping them. We can in fact formally prove this statement:

Construction 4.5. Let $X = A \times B$ denote the disjoint union of $|B|$ copies of A. We identify the two allowed intuitive operations as permutations in $\mathrm{Sym}(X)$:

- Independent left multiplication on fibers: $f \cdot (a, x) = (f(x)a, x)$ for $f \in A^B$.
- Permutation of the fibers: $b \cdot (a, x) = (a, bx)$ for $b \in B$.

Let $G \leq \mathrm{Sym}(X)$ be the group generated by these operations. We define a map $\varphi : A \wr B \to G$ by $fb \mapsto f \circ b$, yielding the action:

$$(fb) \cdot (a, x) = (f(bx)a, bx)$$

Direct computation verifies that φ is a valid homomorphism respecting the semidirect product structure $(fb)(f'b') = f(f')^b bb'$. Furthermore, the action is faithful: if $(f(bx)a, bx) = (a, x)$ for all $(a, x) \in X$, then $b = 1_B$ and $f(x)a = a$ implies $f = 1_{A^B}$. Since φ is an injective homomorphism surjecting onto the generators of G, it is an isomorphism. Thus, $A \wr B \cong G$, proving the wreath product is exactly this group of actions.

The motivation for this construction is that every extension of B by A turns out to be a subgroup of the wreath product. Therefore we can say things about every extension by understanding the wreath product.

Theorem 4.6 (Kaloujnine–Krasner Embedding Theorem)**.** *If E fits into the exact sequence of finite groups*

$$1 \to A \xrightarrow{\iota} E \xrightarrow{\pi} B \to 1,$$

then E is isomorphic to a subgroup of $W := A \wr B$.

Proof. For simplicity, we will consider A as a subgroup of E. Define a set-theoretic section $\tau : B \to E$ such that $\pi \circ \tau = \mathrm{Id}_B$. This is called a *coset representative function*.

By choosing a representative of each coset, we may identify an element $g \in E$ by two things:

- The coset $\pi(g) \in B$ that g lives in.
- The unique displacement $a \in A$ with $g = \tau(\pi(g)) \cdot a$ from the representative of that coset.

Equivalently, every element E can be written uniquely as $\tau(b) \cdot a$ for some $b \in B$ and $a \in A$. This identifies E set-theoretically with $A \times B$. Therefore, the action of E on itself by left multiplication induces an action of E on $A \times B$.

We seek to compute the action of some $g \in E$ on $(a, b) \in A \times B$. Since (a, b) corresponds to the element $\tau(b)a \in E$, we must write $g\tau(b)a$ in $A \times B$ coordinates. By the properties of π

$$\pi(g\tau(b)a) = \pi(g)b$$

so the B coordinate is $\pi(g)b$. Note that

$$\tau(\pi(g)b)a' = g\tau(b)a \qquad \Longrightarrow \qquad a' = \tau(\pi(g)b)^{-1} g\tau(b)a$$

giving us the A coordinate. Therefore the action is

$$g(a, b) = (f_g(b) \cdot a, \pi(g) \cdot b)$$

where $f_g : B \to A$ is defined by:

$$f_g(b) := \tau(\pi(g)b)^{-1} g\tau(b)$$

This expresses that the coset is changed as expected, and the coset representative is twisted by some element of A which depends on g and b. Therefore the action of g on these coordinates can be specified exactly by the data of an element of the wreath product:

$$\phi : E \to W \qquad g \mapsto \pi(g) f_g \tag{1}$$

Since this is the action of E on itself by left multiplication, ϕ is a homomorphism. Since the action of E on itself by left multiplication is faithful, ϕ is injective. Therefore this identification embeds E into the wreath product. □

Lemma 4.7. *Using the notation of Theorem 4.6, if $\tau_1, \tau_2 : B \to E$ are coset representative functions and $\phi_1, \phi_2 : E \to W$ are defined as in Eq. (1) using τ_1 and τ_2 respectively, then ϕ_1 and ϕ_2 are conjugate.*

Proof. The two coset representative functions τ_1 and τ_2 give two different coordinate systems on the same set E. For each $b \in B$, the two chosen representatives $\tau_1(b)$ and $\tau_2(b)$ lie in the same coset, so they differ by a unique element $f(b) \in A$:

$$\tau_2(b) = \tau_1(b) f(b)$$

Consequently

$$\tau_2(b)a = \tau_1(b)(f(b)a)$$

and changing from the τ_2-coordinates to the τ_1-coordinates changes:

$$(a, b) \longmapsto (f(b)a, b)$$

This coordinate change is exactly the action of the element $f1_B \in A \wr B$: it leaves the B-coordinate fixed and translates the copy of A over each $b \in B$ by $f(b)$.

The maps ϕ_1 and ϕ_2 describe the same action of E on itself, but in these two different coordinate systems. Actions written in two coordinate systems differ by conjugation by the coordinate change. Therefore

$$\phi_1(g) = (f1_B)\phi_2(g)(f1_B)^{-1}$$

as desired. □

We have now proven all the necessary properties of the wreath product. Before we prove Schur-Zassenhaus, I introduce the following result.

Theorem 4.8 (Generalized Maschke's Theorem)**.** *If V is a representation of G over some field whose characteristic does not divide $|G|$, and W is a subrepresentation of V, then V has a subrepresentation Z such that $V = W \oplus Z$.* [1]

The proof of this statement is nearly identical to our proof over $\mathbb{C}$. The reader is encouraged to copy that proof and try to see where the fact that the characteristic does not divide $|G|$ is needed.

Now we have all the machinery we need to prove Theorem 4.3.

Proof of Theorem 4.3. We identify A with its image $\iota(A) \leq E$ and write

$$V = A^B \qquad W = A \wr B = V \rtimes B.$$

Let $\phi : E \hookrightarrow W$ be the embedding constructed in Theorem 4.6, and let:

$$U = \phi(A) \leq V$$

Because A is an elementary abelian p-group, it is naturally a vector space over $\mathbb{F}_p$. Consequently, $V = A^B$ is also an $\mathbb{F}_p$-vector space, with its vector-space operations defined pointwise. The action of B on V permutes the coordinates and is therefore $\mathbb{F}_p$-linear. In particular V is a representation of B.

We claim that U is a B-subrepresentation of V. Firstly, U is normal in $\phi(E)$ because A is normal in E. Moreover, $\phi(E)$ projects onto B. Since V is abelian, conjugation on V by an element of $\phi(E)$ depends only on its image in B. Therefore U is a normal subgroup of W. Since the action of B on V is realized by conjugation in the wreath product, it follows that U is preserved by the action of B.

Since $p \nmid |B|$, the generalized Maschke theorem gives a B-subrepresentation $C \leq V$ such that:

$$V = U \oplus C$$

In multiplicative group notation, this says:

$$V = UC \qquad \text{and} \qquad U \cap C = \{1\}$$

Because C is preserved by B, it is a normal subgroup of $W = V \rtimes B$. Let $\psi : W \to W/C$ be the quotient map. We claim that the composition

$$\theta = \psi \circ \phi : E \longrightarrow W/C$$

is an isomorphism. First, θ is injective. If $\theta(g) = 1$, then $\phi(g) \in C$. Since $C \leq V$, this implies

$$\phi(g) \in \phi(E) \cap V = \phi(A) = U \implies \phi(g) \in U \cap C = \{1\}$$

so $g = 1$. On the other hand:

$$|W/C| = |V/C||B| = |U||B| = |A||B| = |E|$$

Therefore θ is an isomorphism. Now let B_0 denote the standard copy of B in W. Since

$$W/C \cong (V/C) \rtimes B_0 \cong U \rtimes B_0,$$

the subgroup $\psi(B_0)$ is a complement to $\psi(U)$ in W/C. Transporting this subgroup across the isomorphism θ gives a subgroup

$$H = \theta^{-1}\big(\psi(B_0)\big) \leq E$$

such that:

$$E = AH \qquad A \cap H = \{1\} \qquad H \cong B$$

Hence $\pi|_H : H \to B$ is an isomorphism, and its inverse is a splitting map $\rho : B \to E$. This proves that the exact sequence splits.

We now prove the conjugacy statement. Let $\rho_i : B \to E$ for $i \in \{1, 2\}$ be two different splitting maps, and use $\tau_i = \rho_i$ as the coset representative function in the construction of Theorem 4.6. Let $\phi_i, \psi_i, \theta_i, C_i$ be constructed as above. Because each ρ_i is a homomorphism, the displacement function associated with an element $\rho_i(b) \in H_i$ is the constant identity function. Consequently $\phi_i(H_i) = B_0$ so by identifying B with B_0 we can write $\rho_i = \phi_i^{-1}|_{B_0}$. By Lemma 4.7

$$\phi_1(g) = c\phi_2(g)c^{-1} \qquad \text{for all } g \in E.$$

so:

$$\rho_1(b) = \phi_1^{-1}(b) = \phi_2^{-1}(c^{-1}bc)$$

Passing to the quotient by C_2 yields:

$$\rho_1(b) = \theta_2^{-1}(\psi_2(c)^{-1}b\psi_2(c))$$

Since $\theta_2 : E \to W/C_2$ is an isomorphism, there is some $g \in E$ with $\theta_2(g) = \psi_2(c)$. Therefore

$$\rho_1(b) = \theta_2^{-1}(\theta_2(g)^{-1}b\theta_2(g)) = g^{-1}\theta_2^{-1}(b)g = g^{-1}\rho_2(b)g$$

so the splittings are conjugate as desired. □

5. Hall's Theorems

Similar to how Sylow theorems guarantee the existence of subgroups of prime power order, Hall's theorems guarantee the existence of subgroups of certain orders under the additional hypothesis that the group is solvable. In particular, Hall's theorems extend the Sylow theorems from a single prime p to an arbitrary set of primes π. Their proof makes use of Burnside's theorem as well as Schur-Zassenhaus, however there is a much more group theoretic flavor to these proofs.

For the remainder, a *minimal criminal* is a group of minimal order satisfying the conditions of the theorem but not the statement. We will use minimal criminal arguments extensively in these sections. The technique is equivalent to induction on the order of G.

I will also use H^g to denote gHg^{-1} and (a, b) to denote the greatest common divisor of a and b.

5.1. **First Theorem.**

We wish to establish a sufficient condition for a group to be solvable. We do this by combining Burnside's theorem with a group-theoretic argument.

Definition 5.1. We denote $\pi(G)$ to be the set of primes dividing the order of G.

Definition 5.2. Let $\pi \subseteq \pi(G)$ be a set of primes dividing the order of G.

- A subgroup $H \leq G$ is a *π-subgroup* if $\pi(H) \subseteq \pi$.
- A subgroup $H \leq G$ is a *Hall π-subgroup* if H is a π subgroup and no prime in π divides the index $[G : H]$.

Lemma 5.3. *Suppose H and K are subgroups of G and $([G : H], [G : K]) = 1$. Then $G = HK$ and $[G : H \cap K] = [G : K][G : H]$.*

Proof. Note that:

$$|HK| = \frac{|H||K|}{|H \cap K|} = |G|\frac{[G : H \cap K]}{[G : H][G : K]}$$

Because the two indices are coprime, their product divides $[G : H \cap K]$, so $|HK| \geq |G|$. Hence equality holds and $|HK| = |G|$, giving the index formula. □

Lemma 5.4. *Suppose π_1, π_2 are two subsets of $\pi(G)$ such that $\pi_1 \cup \pi_2 = \pi(G)$. Then if H_1, H_2 are Hall π_1, π_2-subgroups respectively, $H_1 \cap H_2$ is a Hall $(\pi_1 \cap \pi_2)$-subgroup.*

Proof. By the condition that $\pi_1 \cup \pi_2$ contains every prime dividing G, we see that $[G : H_1]$ is coprime to $[G : H_2]$. Therefore by Lemma 5.3

$$[G : H_1 \cap H_2] = [G : H_1][G : H_2] \implies |H_1 \cap H_2| = |H_1||H_2|/|G|$$

which is exactly the order of a Hall $\pi_1 \cap \pi_2$-subgroup. □

Definition 5.5. We say that a group G has the *Hall property* if for every n dividing the order of G with $(n, |G|/n) = 1$, G has a subgroup of order n.

We will eventually prove that having the Hall property of Definition 5.5 is equivalent to being solvable.

Corollary 5.6. *If G has the Hall property then all Hall subgroups of G have the Hall property.*

Proof. Suppose that $\pi' \subseteq \pi \subseteq \pi(G)$ and H is a Hall π-subgroup of G. We wish to show that H contains a Hall π'-group. Denote

$$\pi_0 = \pi' \cup (\pi(G) \setminus \pi)$$

to be the set of primes which are either not contained in π or contained in π'. Take K to be a Hall π_0-subgroup. By construction $\pi_0 \cup \pi = \pi(G)$. Therefore by Lemma 5.4 $K \cap H$ is a Hall π'-subgroup contained in H as desired. □

Lemma 5.7. *Suppose that G is solvable and let N be a minimal normal subgroup of G. Then N is an elementary abelian p-group for some prime p.*

Proof. Since G is solvable, its subgroup N is also solvable, meaning its commutator subgroup N' is a proper subgroup ($N' < N$). Because N' is characteristic in N and $N \lhd G$, it follows that $N' \lhd G$. The minimality of N combined with $N' < N$ forces $N' = 1$, proving N is abelian. Since N is non-trivial, choose a prime p dividing $|N|$ and define $N[p] = \{x \in N \mid x^p = 1\}$. Because N is abelian, $N[p]$ is a non-trivial characteristic subgroup of N, and therefore $N[p] \lhd G$. By the minimality of N, we must have $N[p] = N$. Thus, every non-identity element of N has order p, meaning N is an elementary abelian p-group. □

Lemma 5.8. *If $H, K, L \subseteq G$ are solvable and $[G : H]$, $[G : K]$, and $[G : L]$ are pairwise coprime, then G is solvable.*

Proof. Suppose that G is a minimal criminal. We claim that G must have a solvable normal subgroup N. Assuming the existence of N let $\phi : G \to G/N$ be the standard quotient map and let $\overline{G}, \overline{H}, \overline{K}, \overline{L}$ be the images of G, H, K, L under this map. Note that $\overline{G}, \overline{H}, \overline{K}, \overline{L}$ satisfy the conditions of the theorem and therefore by minimality of G, $\overline{G}$ is solvable. Then because G/N and N are solvable G must be solvable, a contradiction.

Now we prove the existence of N. Let M be a minimal normal subgroup of H and note that M is an elementary abelian p-group for some prime p. Since p cannot divide both $[G : K]$ and $[G : L]$ we suppose without loss of generality that p does not divide $[G : K]$. By the first Sylow theorem K contains a Sylow p-subgroup P. Since P is also Sylow in G, the second Sylow theorem gives some $g \in G$ such that:

$$M \subseteq P^g \implies M \subseteq K^g$$

By Lemma 5.3 $G = K^g H$. Suppose that $m \in M$ and note for any $kh \in G$ with $k \in K^g$ and $h \in H$ we have

$$m^{kh} = k(hmh^{-1})k^{-1} \in kMk^{-1} \subseteq K^g$$

by normality of M in H. Therefore all the conjugates of M are contained in K^g. We let N be the normal subgroup generated by M. By the preceding argument $N \subseteq K^g$ and is therefore a proper normal subgroup. N is solvable since it is a subgroup of the solvable group K^g. □

Theorem 5.9. *If G has the Hall property then G is solvable.*

Proof. Suppose that $|G| = p_1^{\alpha_1} \dots p_n^{\alpha_n}$ is a minimal criminal. If $n \leq 2$ then G is solvable by Burnside's theorem. Otherwise let H, K, L be Hall subgroups of G with indices $p_1^{\alpha_1}, p_2^{\alpha_2}, p_3^{\alpha_3}$ respectively. By Corollary 5.6 each of these subgroups has the Hall property so by minimality of G they are all solvable. However they have coprime indices in G, so by Lemma 5.8 G is solvable, a contradiction. □

5.2. **Second Theorem.** Now that we have proven the first Hall theorem, we seek to prove the converse. This will establish remarkable necessary and sufficient conditions for a group to be solvable.

Theorem 5.10. *If G is solvable, then it has the Hall property.*

Proof. Suppose that G is a minimal criminal and let $\pi \subseteq \pi(G)$ be such that no Hall π-subgroup exists. Let M be a minimal normal subgroup of G and note that M is an elementary abelian p-group for some prime p. There are two cases:

$p \in \pi$: By minimality of G, G/M contains a Hall π-subgroup $\overline{H}$. The lift of $\overline{H}$ into G is a Hall π-subgroup of G, giving a contradiction.

$p \notin \pi$: Let $\overline{H}$ be a Hall π-subgroup of G/M and let H be its lift into G. Since $|M|$ and $[H : M]$ are coprime Schur-Zassenhaus gives a complement K of M in H. By construction K is a Hall π-subgroup, a contradiction.

□

5.3. **Third Theorem.** The final Hall theorem is analogous to the result that all Sylow p-subgroups are conjugate to each other. It states that Hall π-subgroups are conjugate and furthermore any π-subgroup is contained in a Hall π-subgroup. This result gives us a great level of understanding of the structure of solvable groups.

Theorem 5.11. *Suppose that G is a solvable group and $\pi \subseteq \pi(G)$. If H is a π-subgroup and K is a Hall π-subgroup, then there is some $g \in G$ such that $H \subseteq K^g$.*

Proof. Suppose that G with subgroups H and K is a minimal criminal. Let M be a minimal normal subgroup of G and note it must be an elementary abelian p-group for some prime p. Then $\overline{K} = KM/M$ is a Hall π-subgroup of $\overline{G} = G/M$ and $\overline{H} = HM/M$ is a π-subgroup of $\overline{G}$. By minimality of G there is some $g \in G$ such that $\overline{H} \subseteq \overline{K}^g$. Taking the lifts we have $HM \subseteq K^g M$. There are two cases:

$p \in \pi$: $|K^g M|$ must equal K^g since otherwise $|K^g M|$ is divisible by a higher power of p than K. Therefore $HM \subseteq K^g$ and $H \subseteq K^g$.

$p \notin \pi$: Note that

$$(H)M \subseteq (K^g \cap (HM))M \subseteq (HM)M = HM$$

so H and $K^g \cap (HM)$ are both complements to M in HM. Then by Schur-Zassenhaus they are conjugate in HM. Hence for $x \in HM$ we have $H = (K^g \cap (HM))^x \subseteq K^{xg}$.

□

Corollary 5.12. *Suppose that G is solvable. For a given set of primes $\pi \subseteq \pi(G)$, all Hall π-subgroups are conjugate.*

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